EXT OF FINITE PROJECTIVE MODULES WITH CONTINUOUS GROUP ACTION AND GROUP COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Continuous cohomology of a topological group with coefficients in a module. Same chapter, the cohomology H^i(G, M) of a topological group with coefficients in a discrete module, and the standard resolution by continuous functions. Supplies the right hand side of the asserted identification.
- Ext groups of R-G-modules and finite projectivity of M over R. Same chapter, the Ext groups in the category of R-G-modules and the use of the dual module M^vee = Hom_R(M, R). Supplies the left hand side and the computation by adjunction.

Source reference. Original lemma 00DE: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8923–8923. Navigation pointer invoked at proof line 8923; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0DVB: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8924–8924. Navigation pointer invoked at proof line 8924; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let G be a topological group. Let R be a ring. Let M, N be R-G-modules. If M is finite projective as an R-module, then $Ext^i(M, N) = H^i(G, M^\vee \otimes_R N)$ (for notation see proof).*

Proof. The module $M^\vee = \mathrm{Hom}_R(M, R)$ endowed with the contragredient action of G. Namely $(g \cdot \lambda)(m) = \lambda(g^{-1} \cdot m)$ for $g \in G$, $\lambda \in M^\vee$, $m \in M$. The action of G on $M^\vee \otimes_R N$ is the diagonal one, i.e., given by $g \cdot (\lambda \otimes n) = g \cdot \lambda \otimes g \cdot n$. Note that for a third R-G-module E we have $\mathrm{Hom}(E, M^\vee \otimes_R N) = \mathrm{Hom}(M \otimes_R E, N)$. Namely, this is true on the level of R-modules by Algebra, Lemmas 00DE and 0DVB and the definitions of G-actions are chosen such that it remains true for R-G-modules. It follows that $M^\vee \otimes_R N$ is an injective R-G-module if N is an injective R-G-module. Hence if $N \to N^\bullet$ is an injective resolution, then $M^\vee \otimes_R N \to M^\vee \otimes_R N^\bullet$ is an injective resolution. Then

$$\mathrm{Hom}(M, N^\bullet) = \mathrm{Hom}(R, M^\vee \otimes_R N^\bullet) = (M^\vee \otimes_R N^\bullet)^G$$

Since the left hand side computes $\mathrm{Ext}^i(M, N)$ and the right hand side computes $H^i(G, M^\vee \otimes_R N)$ the proof is complete. □